David Mohaisen, Professor
University of Central Florida
Department of Computer Science
Cybersecurity and Privacy Cluster

https://cs.ucf.edu/~mohaisen/

Neural Network Classifiers and Sequence Labeling

① Classifiers
② Training softmax
③ Neural net classifiers
④ NER
⑤ Window classification
⑥ ANN FF and max margin loss

Lecture Objectives

1. Highlight some logistics
2. Classifiers and training softmax
3. Neural network classifiers
4. Window classification
5. NER and its working as an example

Logistics

1

Modules 2 is released

2

Proposal not graded yet

5

Office hour time; 2:00pm – 3:00pm, Tuesdays
Away: Feb 6 to Feb 12

3

Recordings are posted on YouTube right after the lecture

4

Assignment 1 was due on Jan 26, 2026,
Assignment 2 is out, due Feb 9.

Classifiers Overview

Classifiers: Definitions

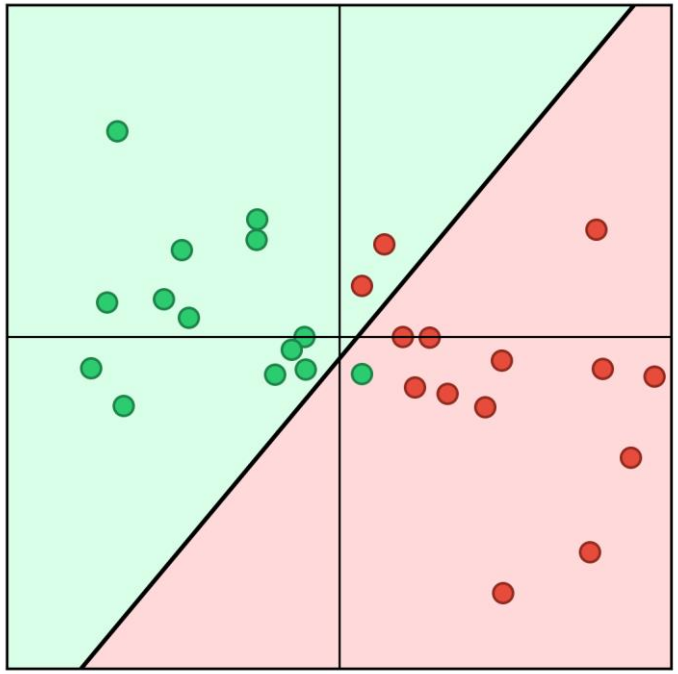
- A training set is given, consisting of samples $\{x_i, y_i\}^n_{i=1}$
  - x_i are inputs, e.g., words (indices or vectors), sentences, documents, etc.
    - Processed using tokenization, embeddings, etc.
  - y_i is labels (one of C classes) we try to predict; examples of classes include:
    - Classes: sentiment, named entity, decisions
    - Decisions: accept, reject, defer, etc.
    - Named entity: person, location, organization, etc.
    - Later: multi-word sequence (e.g., sequence labeling tasks; POS, and language generation tasks; story telling).

Classifiers: Intuition

- Training data $\{x_i, y_i\}^n_{i=1}$
  - x: fixed 2D word vectors to classify: each input is encoded.
  - Using softmax/logistic regression
    - Softmax: multi-class classifier (outputs probabilities for all classes)
    - LR: binary classifier (outputs probabilities for two classes).
  - Linear decision boundaries

$$p(y|x) = \frac{\exp(W_{y\cdot}x)}{\sum_{c=1}^{C}\exp(W_{c\cdot}x)}$$

  - Where W is a weight used to determine decision boundaries; applied to x to calculate scores of y
  - The model training involves learning W.



$$z_j = W_j \cdot x + b_j \qquad P(y=j|x) = \frac{e^{z_j}}{\sum_{k=1}^{C} e^{z_k}} \qquad \text{Loss} = -\sum_{i=1}^{n}\log(P(y_i = j|x_i))$$

Classifiers: Example (Softmax)

- We can tease apart the prediction part as

$$p(y|x) = \frac{\exp(W_{y\cdot}x)}{\sum_{c=1}^{C}\exp(W_{c\cdot}x)}$$

- Take the yth row of W and multiply it with x (repeat that for f_c for $c = 1$ all the way to C (number of classes).

$$W_{y\cdot}x = \sum_{i=1}^{d} W_{yi}x_i = f_y$$

- Apply softmax function to get normalized probabilities

$$p(y|x) = \frac{\exp(f_y)}{\sum_{c=1}^{C}\exp(f_c)} = \text{softmax}(f_y)$$

Training Softmax

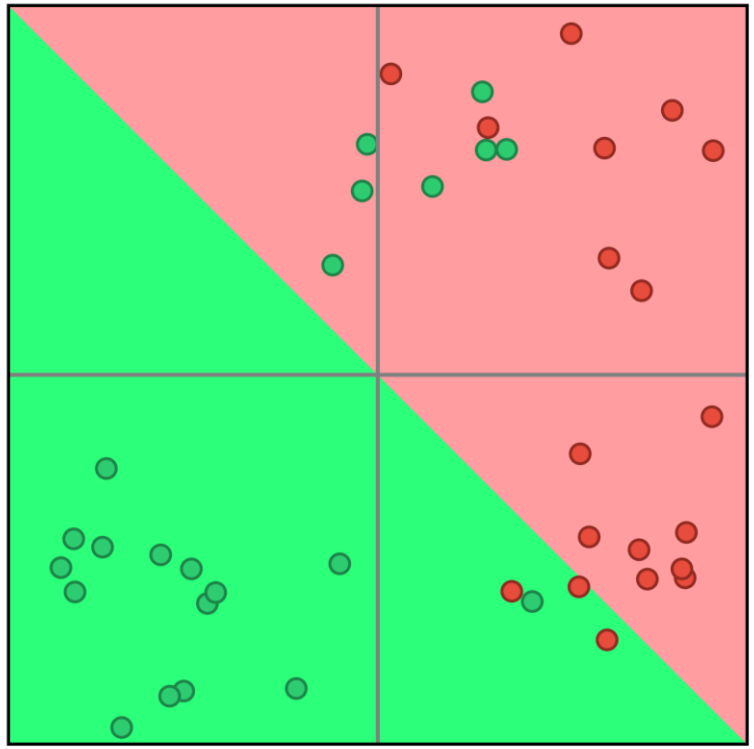
- For each training example (x,y), we want to maximize probability of the correct label y
  - Or we can minimize the negative log probability of that class:

$$-\log p(y|x) = -\log\left(\frac{\exp(f_y)}{\sum_{c=1}^{C}\exp(f_c)}\right)$$

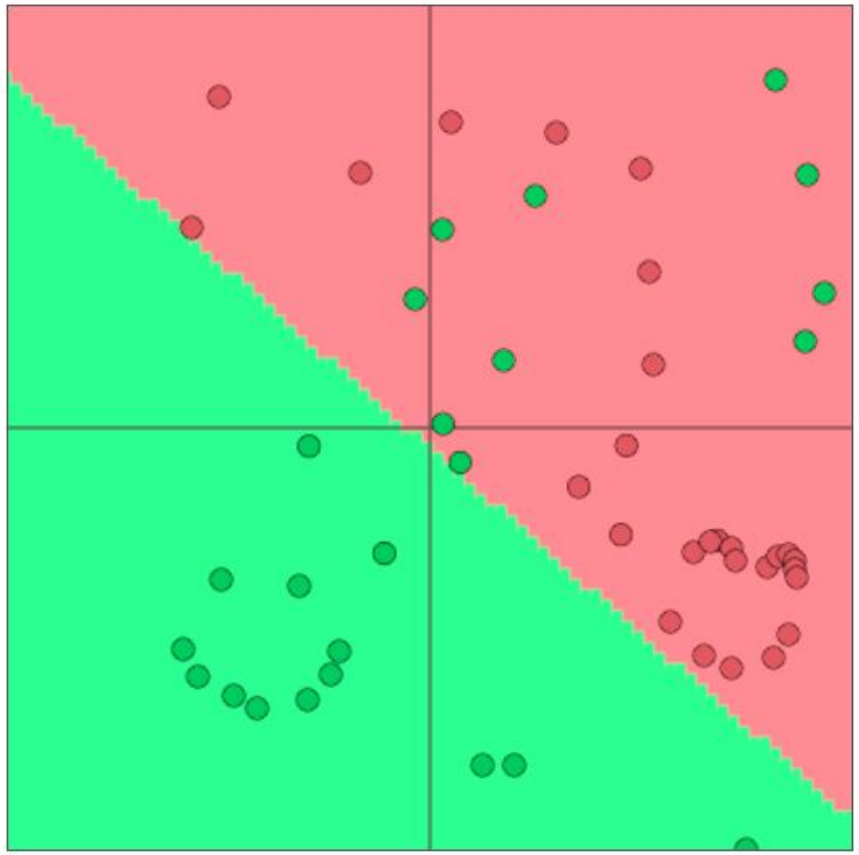
Problems with Softmax and Neural Classifiers

Problems with Softmax

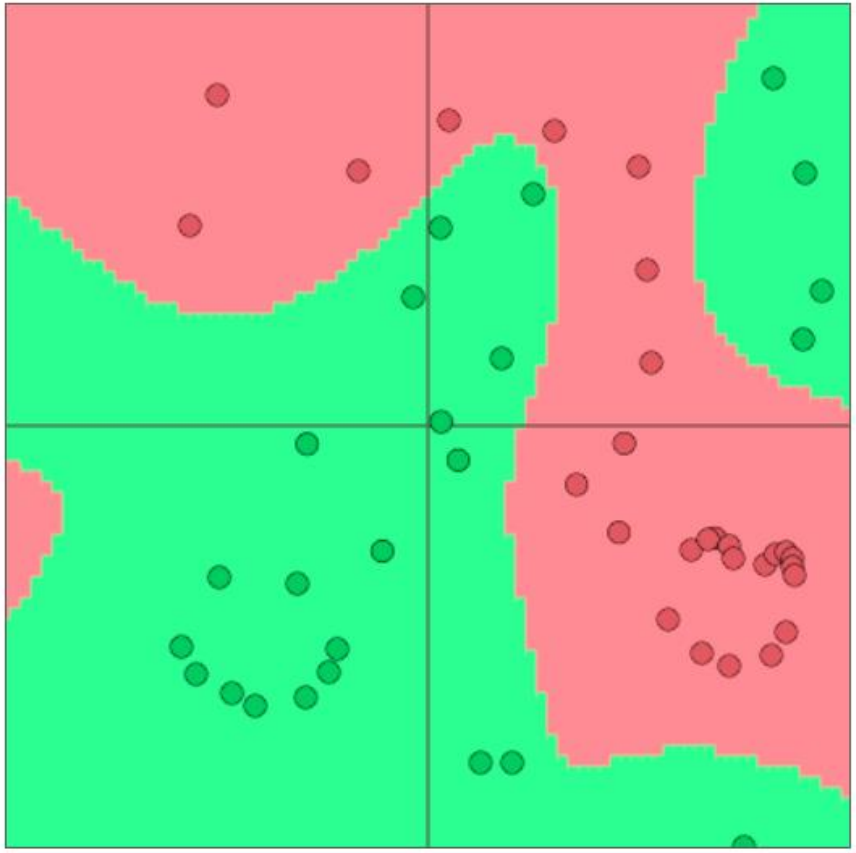
- Softmax is not powerful alone
- Softmax gives only linear boundaries



Neural Networks Classifiers as a Solution



Softmax - LR



Neural Networks - Reality

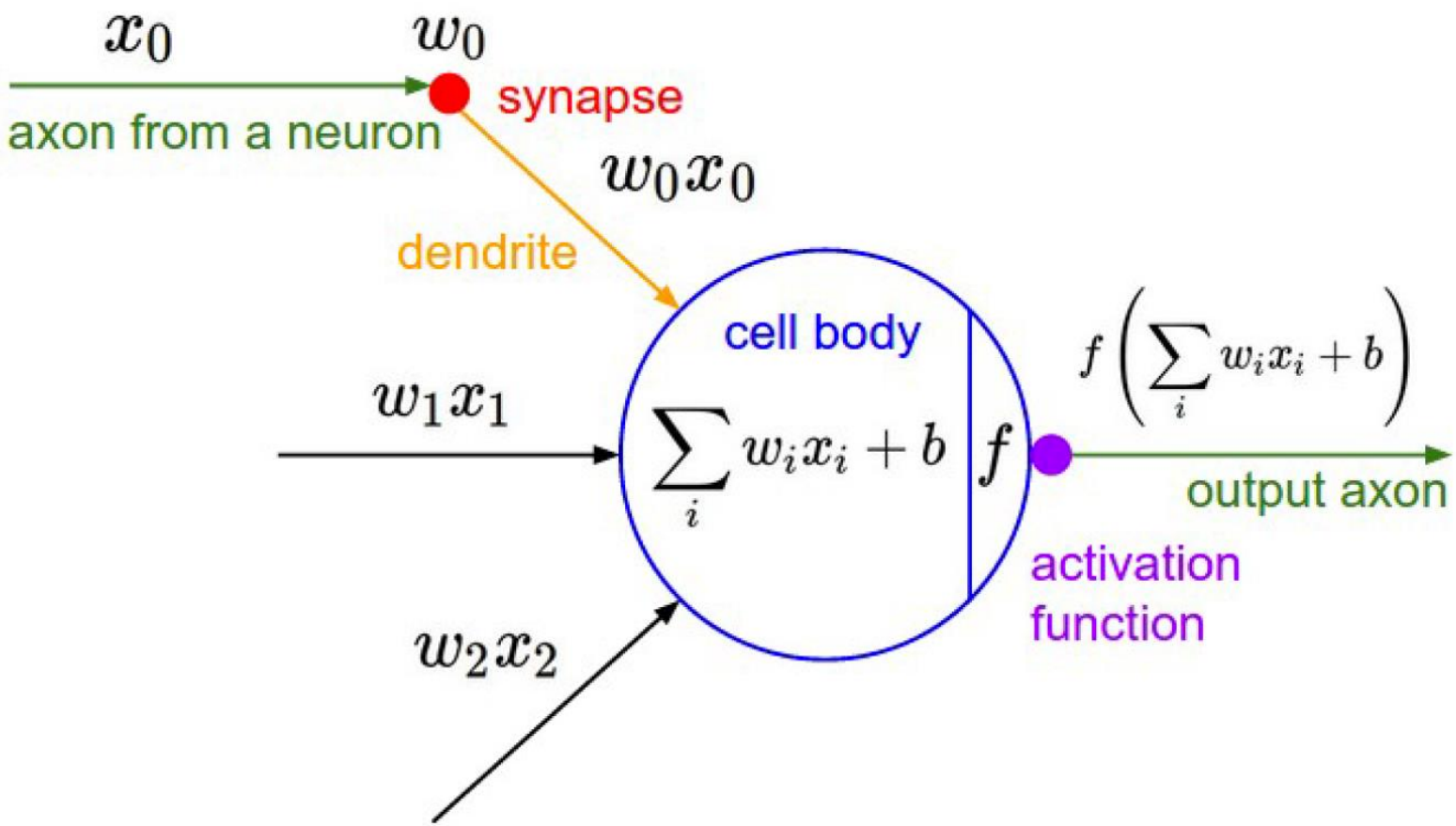
NN Classification Diff with Word Vectors

- Commonly used in NLP deep learning:
  - Learn both W and word vector x
  - Learn both convention parameters and representations
  - The word vectors re-represent one-hot vectors for easy classification with a linear softmax classifier via layer x

$$\nabla_\theta J(\theta) = \begin{bmatrix} \nabla_{W_{\cdot 1}} \\ \vdots \\ \nabla_{W_{\cdot d}} \\ \nabla_{x_{aardvark}} \\ \vdots \\ \nabla_{x_{zebra}} \end{bmatrix} \in \mathbb{R}^{Cd+\boxed{Vd}}$$

Very large number of parameters!

Artificial Neural Network, intro.



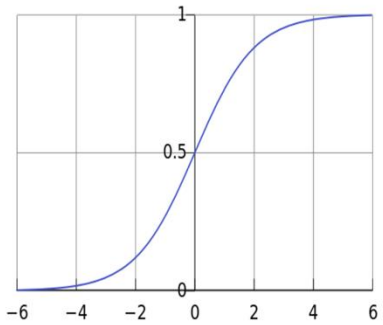
Artificial Neural Network, cont.

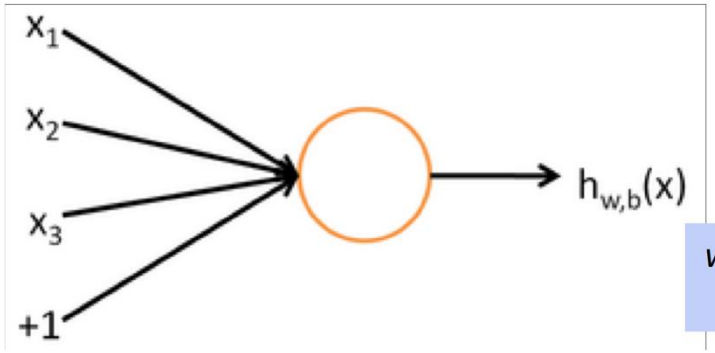
- A neuron can be a binary logistic regression unit
  - *f*=nonlinear activation function, w=weight, b=bias, h=hidden, x=inputs

$$h_{w,b}(x) = f(w^{\mathsf{T}}x + b)$$

b: We can have an "always on" feature, which gives a class prior, or separate it out, as a bias term

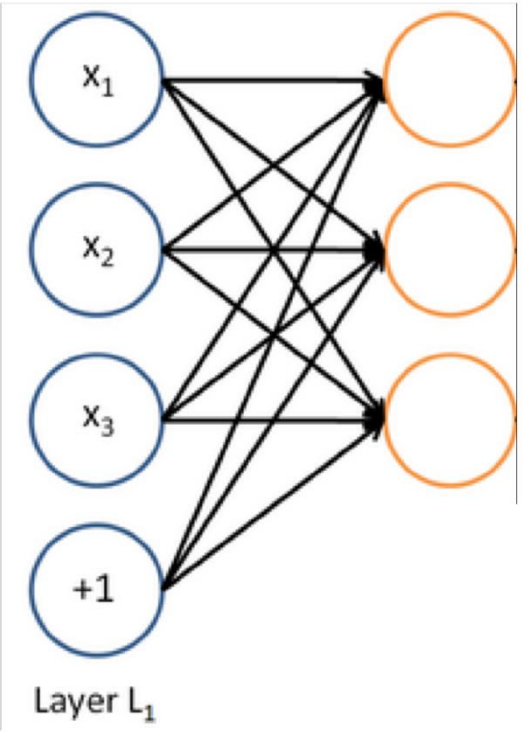
$$f(z) = \frac{1}{1+e^{-z}}$$





w, *b* are the parameters of this neuron i.e., this logistic regression model

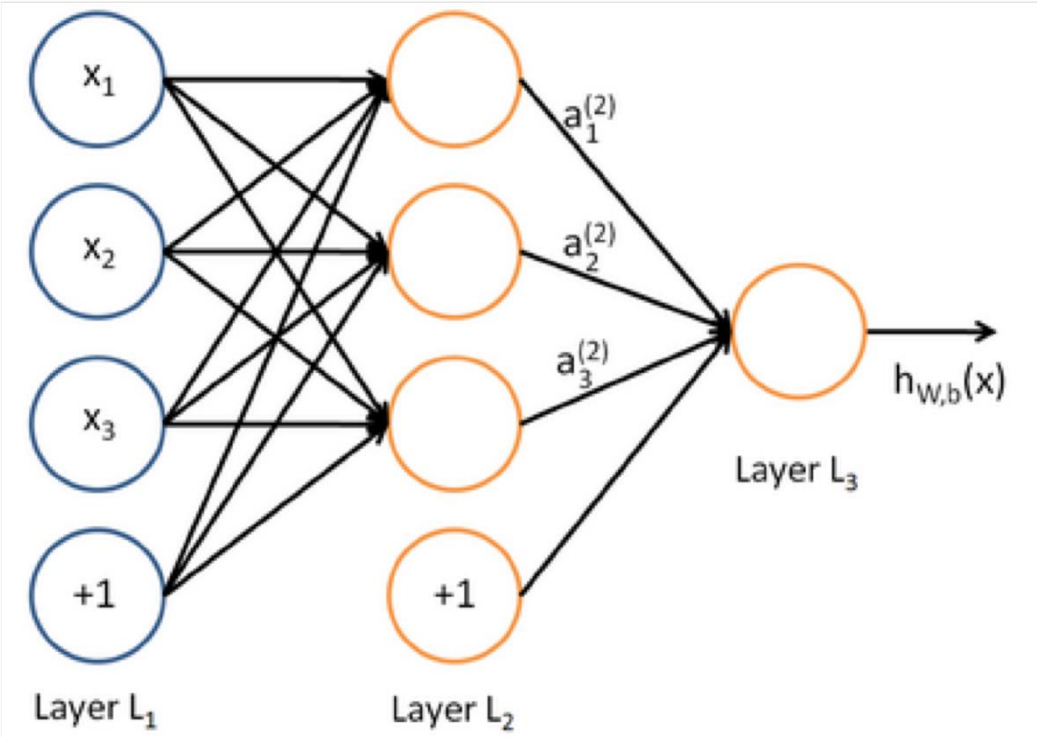
Artificial Neural Network, cont.



But we don't have to decide ahead of time what variables these logistic regressions are trying to predict!

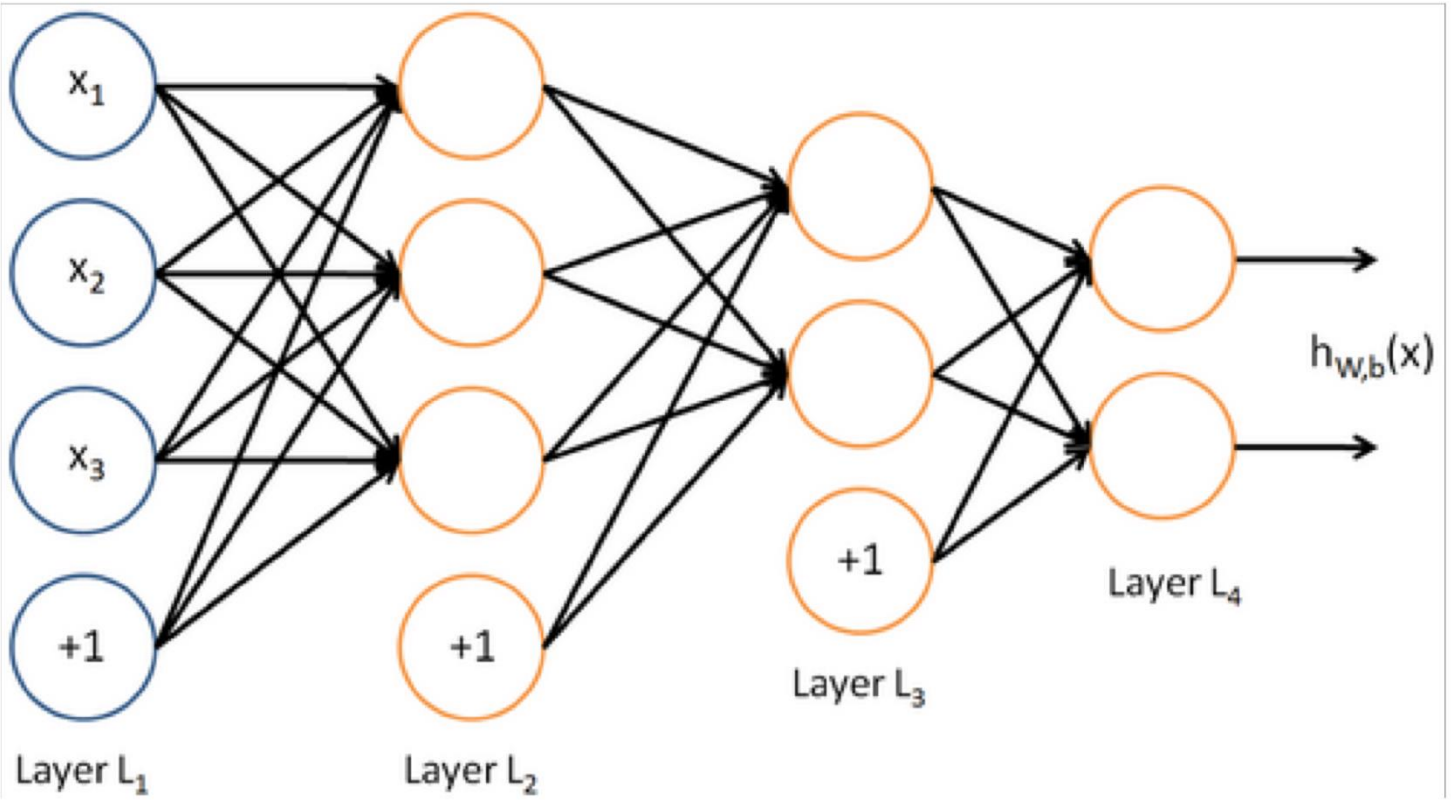
We use a large number of parameters to represent the input; several logistic regressions

Artificial Neural Network, cont.



It is the loss function that will direct what the intermediate hidden variables should be, so as to do a good job at predicting the targets for the next layer, etc.

Artificial Neural Network, cont.



Artificial Neural Network, cont.

- We have

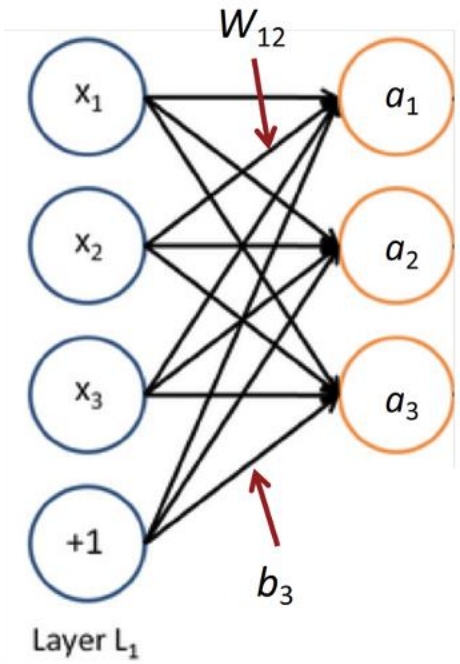
$$a_1 = f(W_{11}x_1 + W_{12}x_2 + W_{13}x_3 + b_1)$$
$$a_2 = f(W_{21}x_1 + W_{22}x_2 + W_{23}x_3 + b_2)$$
$$\text{etc.}$$

- In matrix notation

$$z = Wx + b$$
$$a = f(z)$$

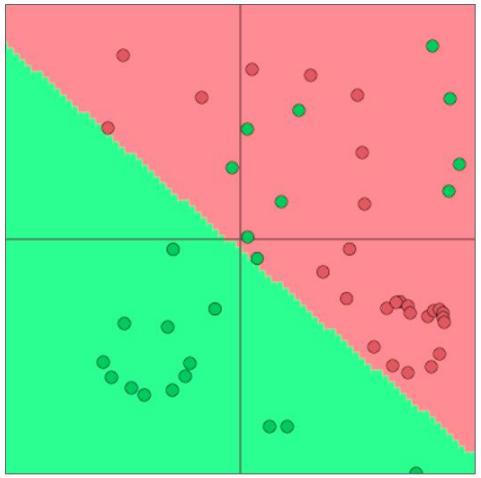
- Activation f is applied element-wise

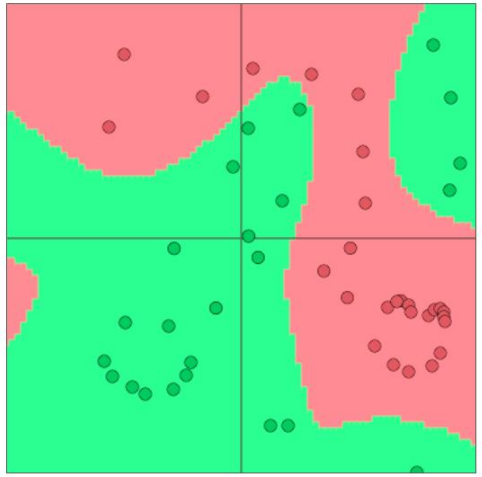
$$f([z_1, z_2, z_3]) = [f(z_1), f(z_2), f(z_3)]$$

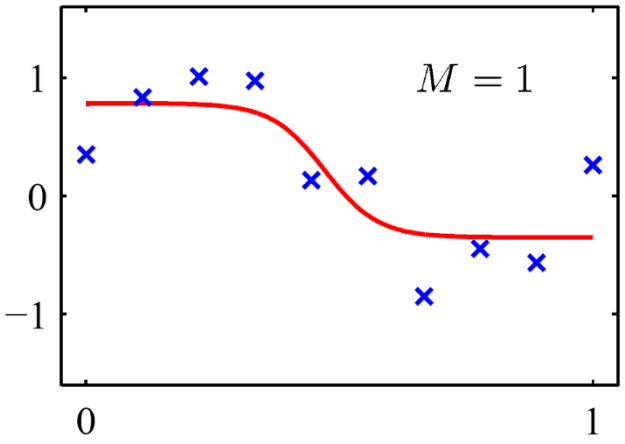


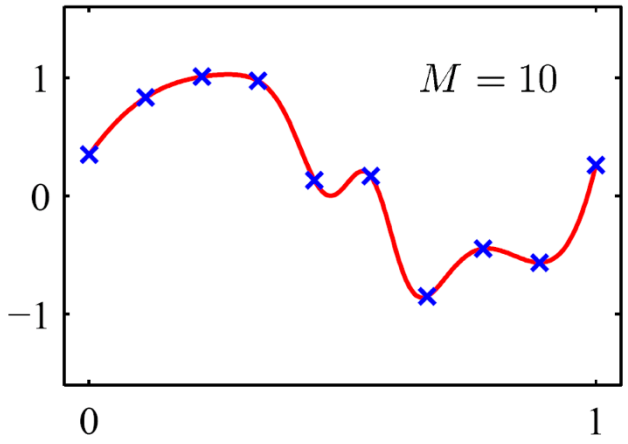
Nonlinear Functions

- Example: function approximation, e.g., regression or classification
  - Without non-linearities, deep neural networks can't do anything more than a linear transform
  - Extra layers compiled down into a single linear transform: W_1; W_2; $x = W_x$
  - With more layers, they can approximate more complex functions!









Named Entity Recognition

Task: Named Entity Recognition (NER)

- The task: find and classify names in text:
  - The European commission [ORG] said on Thursday it disagreed with German [MISC] advice. Only France [LOC] and Britain [LOC] backed Fischler [PER] proposal. What we have to be extremely careful of is how other countries going to take German's lead, Welsh National Farmers' union [ORG] (NFU [ORG]) chairman John Jones [PER] said on BBC [ORG] radio.

- Possible purposes:
  - Tracking mentions of particular entities in documents
  - For question answering, answers are usually named entities
  - A lot of wanted information is really associations between named entities
  - The same techniques can be extended to other slot-filling classifications

NER on Word Sequences

- We predict entities by classifying words in context and then extract entities as word subsequences (using BIO; inside-outside-beginning encoding):

Foreign	ORG	B-ORG
Ministry	ORG	I-ORG
spokesman	O	O
Shen	PER	B-PER
Guofang	PER	I-PER
told	O	O
Reuters	ORG	B-ORG
that	O	O
:	:	👆 BIO encoding

Named Entity Types
ORG → **Organization**
PER → **Person**
O → **Outside** (not part of any named entity)
BIO Encoding Prefixes
B- → **Beginning** of an entity span
I- → **Inside** (continuation) of an entity span
Examples from the figure
B-ORG → Beginning of an **Organization** name
I-ORG → Inside an **Organization** name
B-PER → Beginning of a **Person** name
I-PER → Inside a **Person** name
BIO (overall)
BIO → **Beginning–Inside–Outside** sequence labeling scheme

Why is NER Hard?

- **Determining Entity Boundaries**
  - It can be challenging to identify where an entity begins and ends, especially in complex phrases.
  - Example: "First National Bank"
    - Is it a single entity (organization) or multiple components?

- **Identifying Entities**
  - It’s not always clear if a phrase refers to an entity or just descriptive text.
  - Example: "Future School"
    - Is it a proper noun (entity) or a general term?

Why is NER Hard? cont.

- **Classifying Unknown or New Entities**
  - Novel or unfamiliar entities can be difficult to categorize.
  - Example: "To find out more about Zig Ziglar"
    - Is Zig Ziglar a person (PER) or something else?

- **Ambiguity in Entity Classes**
  - Some entities can belong to multiple categories, depending on context.
  - Example: "Charles Schwab"
    - Is it a person (PER) or an organization (ORG)?

Ambiguity in Single-Word Classification

1. Auto-antonyms (Contronyms)

1. Words with opposite meanings depending on context:
    1. *Sanction*: Can mean **permit** or **punish**.
    2. *Seed*: Can mean **to place seeds** or **to remove seeds**.

2. Resolving Ambiguity in Named Entities

1. Context is critical to identify the correct entity:
    1. *Paris*: Could refer to **Paris, France**, **Paris, Texas**, or **Paris Hilton**.
    2. *Hathaway*: Could refer to **Berkshire Hathaway** (organization) or **Anne Hathaway** (person).

Disambiguation requires understanding context and linking words to their most probable meaning or reference based on surrounding information. Advanced models use context-aware techniques to resolve such ambiguities effectively.

Window Classification

Solution: Window Classification

- **Classifying Words in Context**:
  - The goal is to determine the class of a word (e.g., person, location, organization, none) based on its surrounding words (context window).
  - Example: In "Paris Hilton stayed in Paris," the context helps differentiate between the **person** and the **location**.

Solution: Window Classification; Approach

1. **Using Word Vectors**:
    1. A simple method involves averaging the word vectors within a fixed-size context window around the target word.
    2. The resulting average vector is classified into one of the entity classes.

2. **Challenges with Averaging Word Vectors**:
    1. **Loss of Positional Information**:
        1. Averaging ignores the order of words, which is critical for context understanding.
        2. Example: "Hilton in Paris" vs. "Paris Hilton."
    2. **Contextual Nuance**:
        1. Subtle changes in meaning from word order or relationships are not captured.

Window Classification: softmax

- Train softmax classifier to classify a center word by taking concatenation of word vectors surrounding it in a window.
  - Ex.: Classify Paris in the context of this sentence with a window of length 2:

… museums in Paris are amazing … .

$$X_{window} = [\ x_{museums} \quad x_{in} \quad x_{Paris} \quad x_{are} \quad x_{amazing}\ ]^T$$

- Resulting vector $x_{window} = \boxed{x \in R^{5d}}$, a column vector!

Training a Softmax with CW

- **Concept**
  - Train a **softmax classifier** to classify a center word by leveraging its surrounding context.
  - Use the **concatenation of word vectors** from neighboring words in a fixed-size window as input.

- **Example: Classifying "Paris"**
  - **Sentence**: "Paris Hilton stayed in Paris."
  - **Target Word**: "Paris" (center word).
  - **Window Size**: 2 (one word before and one after).
    - **Context Words for First "Paris"**: ["Hilton", "stayed"].
    - **Context Words for Second "Paris"**: ["in", "Paris"].

Training a Softmax with CW, Steps.

1. Input Preparation:

1. Represent each word as a **word vector** (e.g., embeddings).
2. **Concatenate** vectors for the surrounding words within the window: x=[vec(Hilton), vec(stayed)] for the first "Paris".

2. Training:

1. Pass concatenated vectors through a **softmax classifier**.
2. Output probabilities for each class (e.g., **person**, **location**, etc.).

3. Classification:

1. Assign the class with the highest probability to the center word.

Binary Classification for NER Location

- **Sentence Variations**:
  - *Not all museums in Paris are amazing.*
  - *All museums in Paris are amazing.*
  - *Not all museums in Paris.*

- **Context Window Effect**:
  - For "museums," the window size and content affect its interpretation:
    - *Window 1*: ["Not", "all"] suggests **limitation** or **negativity**.
    - *Window 2*: ["All", "amazing"] implies **positivity** or **universality**.
    - *Window 3*: ["Not", "museums", "in"] is **incomplete**, leading to ambiguity.

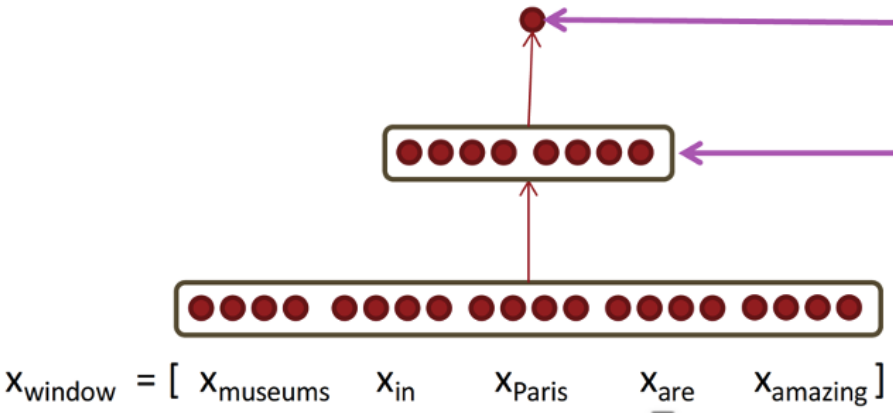
ANN Feed-Forward

- Using neural activation a simply to give an unormalized score

- We compute the wind $score(x) \quad = \quad U^T a \in \mathbb{R}$ neural network
  - s=score(“museum in Paris is amazing)

$$s = U^T f(Wx + b)$$

$$x \in \mathbb{R}^{20\times 1}, W \in \mathbb{R}^{8\times 20}, U \in \mathbb{R}^{8\times 1}$$



$$s = U^T a$$

$$a = f(z)$$

$$z = Wx + b$$

The Max-Margin Loss

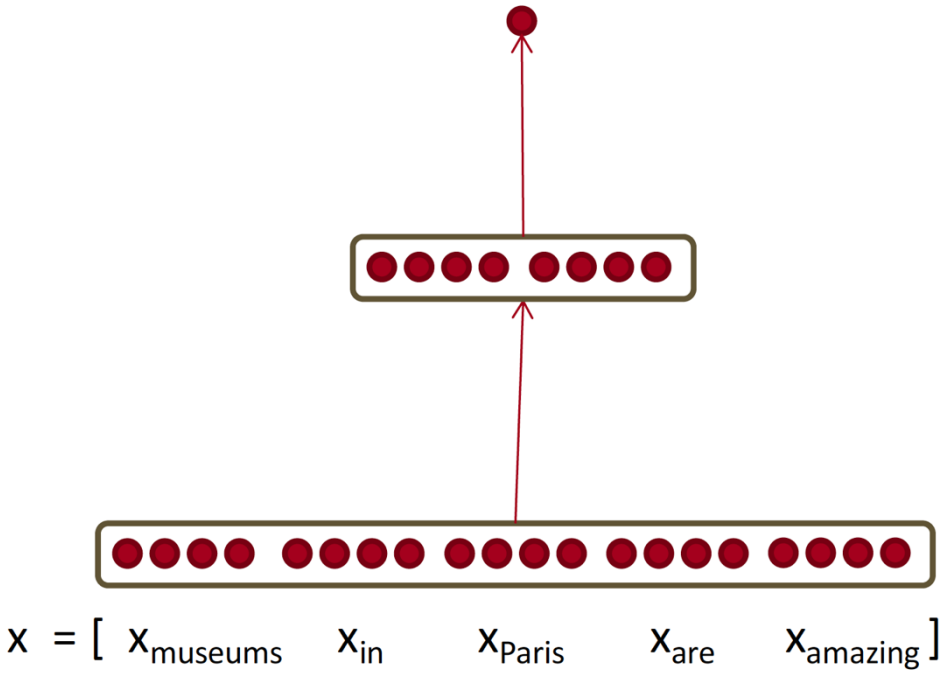
- To improve classification, we aim to increase the score of the true window (S) while lowering the score of the corrupted window (S_c).
  1. **Scores**:
     1. S=score("museum in Paris are amazing").
     2. S_c=score("not all museums in Paris").
  2. **Loss Function**:
     1. Minimize: J=max (0, 1−S+Sc)
     2. Ensures S is higher than S_c by a margin of 1.
  3. **Optimization**:
     1. Use Stochastic Gradient Descent (SGD) to adjust model parameters.
     2. Update only when J>0, i.e., when 1−S+Sc>0.
  - This method separates true and corrupted contexts effectively, improving classification performance.

The Max-Margin Loss

- Each context window with a Named Entity Recognition (NER) **location** at its center should have a score at least **+1 higher** than any corrupted window (i.e., one without a location at its center).
  - For each true window (containing a location entity at the center), sample multiple **corrupted windows** that lack the location entity or have incorrect contexts.

Scoring Illustration

$$s = \boldsymbol{u}^T \boldsymbol{h}$$

$$\boldsymbol{h} = f(\boldsymbol{W}\boldsymbol{x} + \boldsymbol{b})$$

$$\boldsymbol{x} \quad \text{(input)}$$